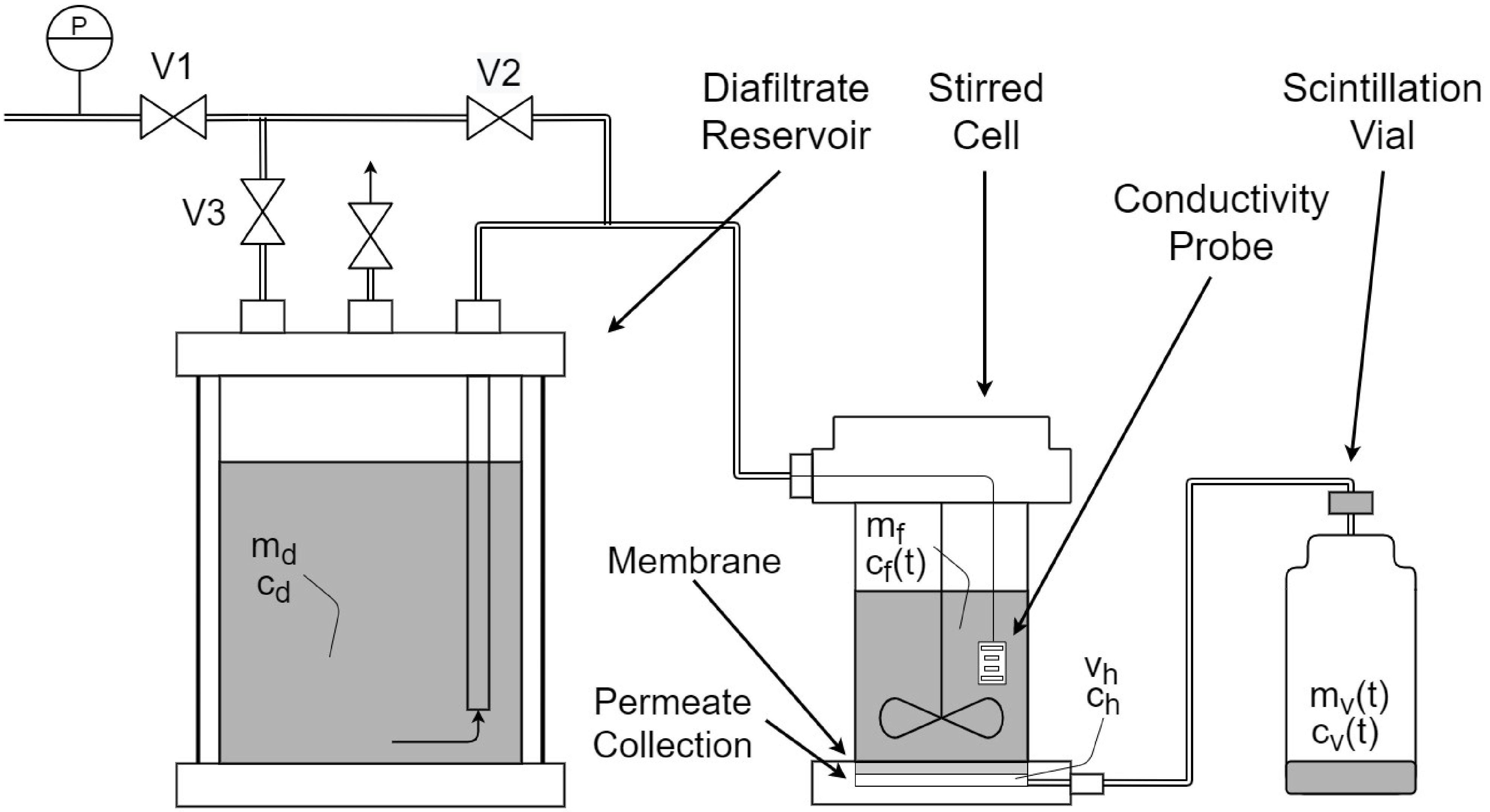
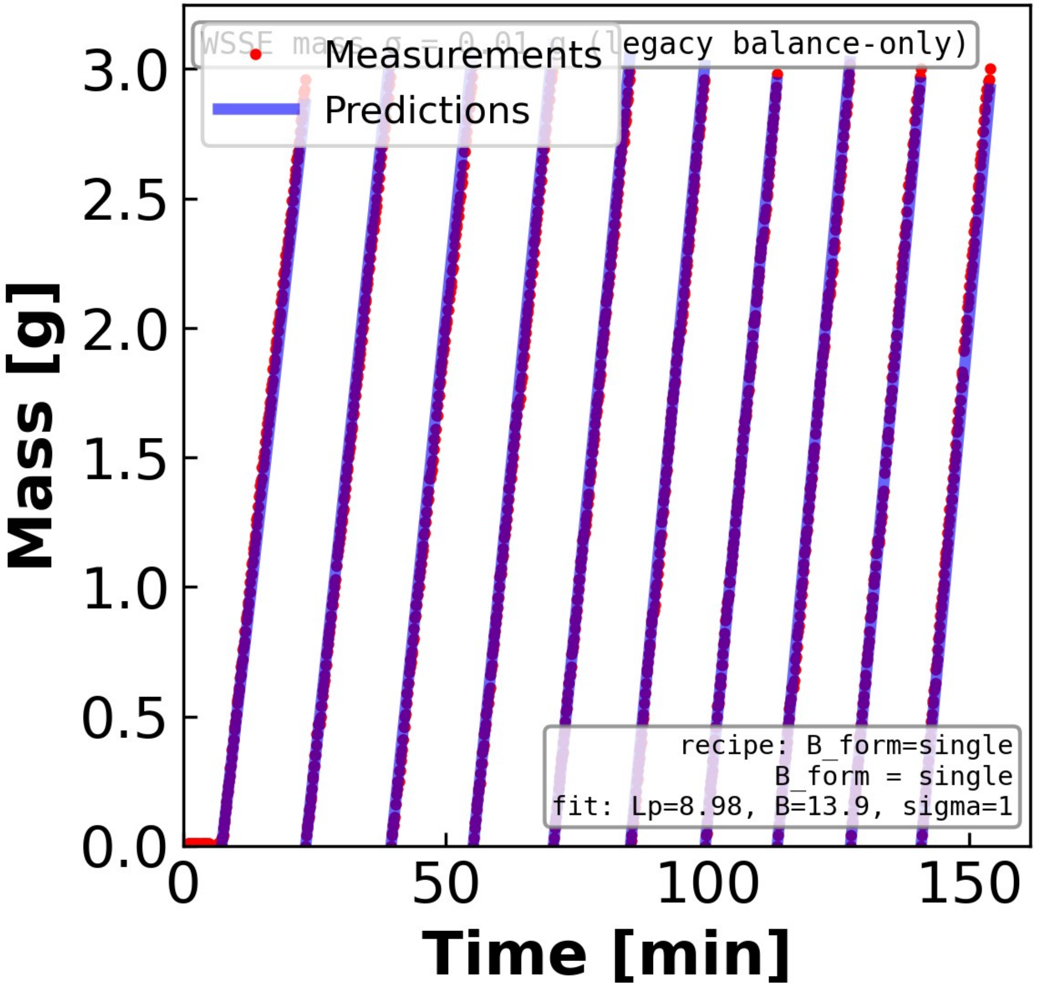
Backup — Supporting Figures

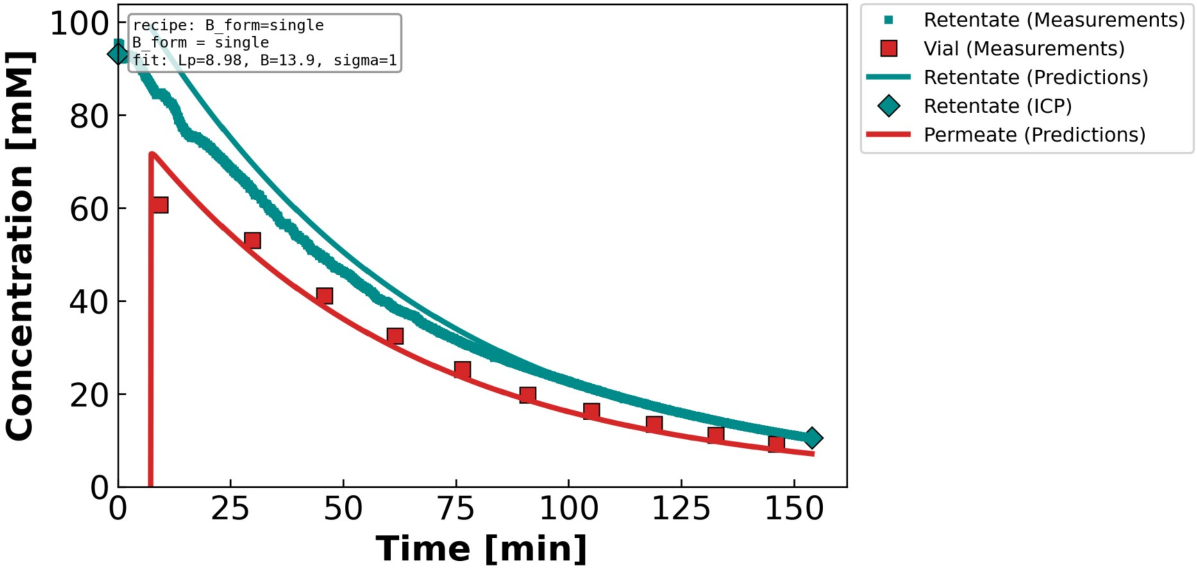
Fits, full σ×B contours, Péclet regression, NaCl B(c), concentration–flux

Backup — Physics-Based Model Links Data to Transport Parameters

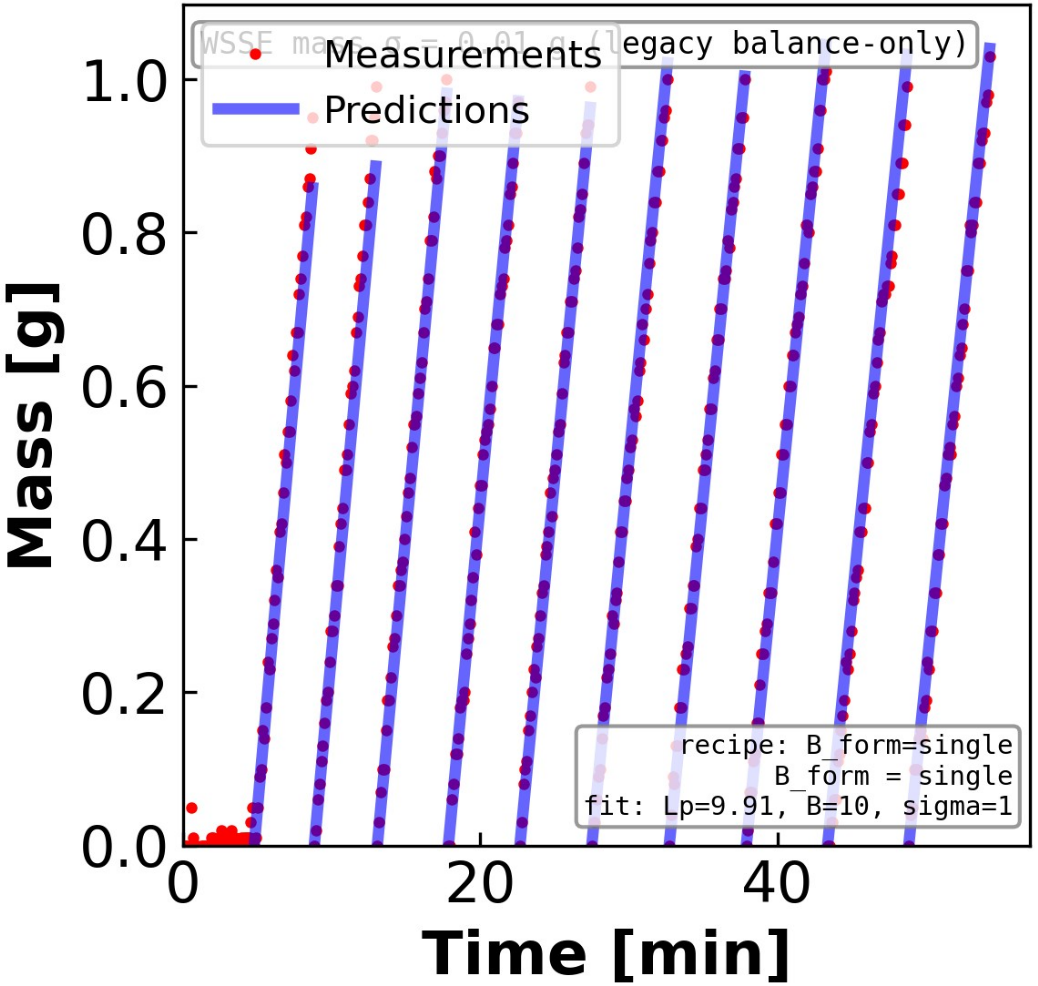


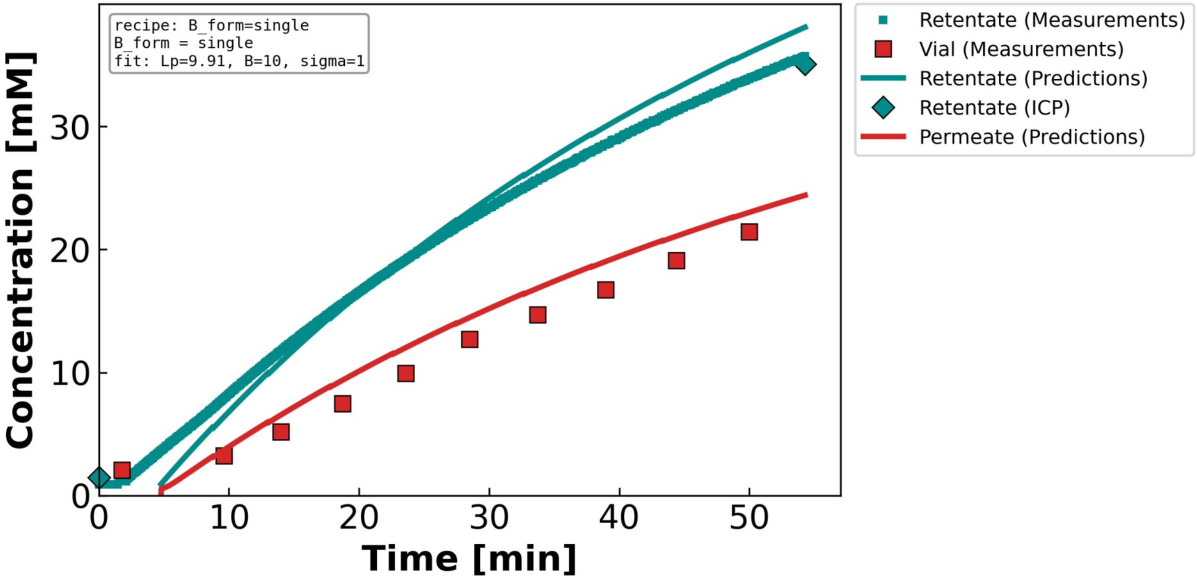
Backup — Figure 1. Measured and model-predicted time profiles for a representative…



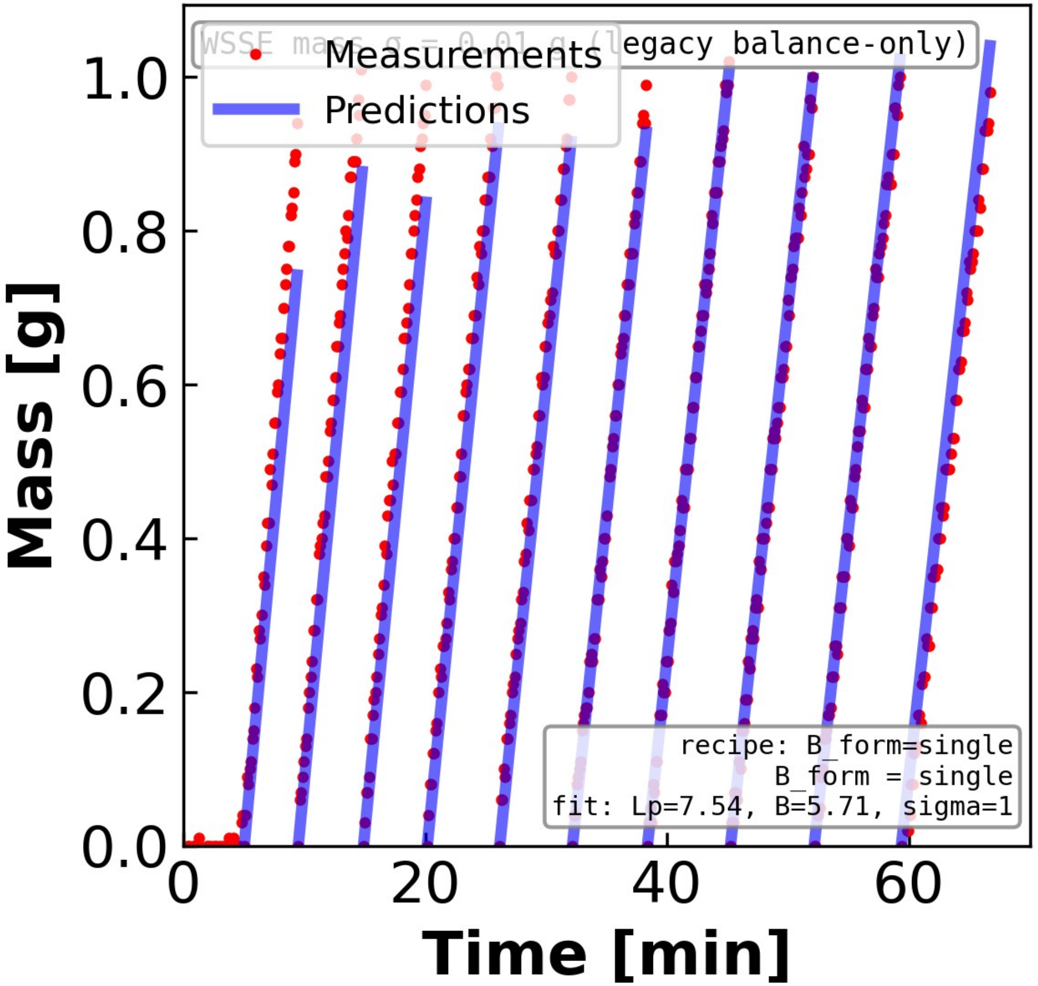


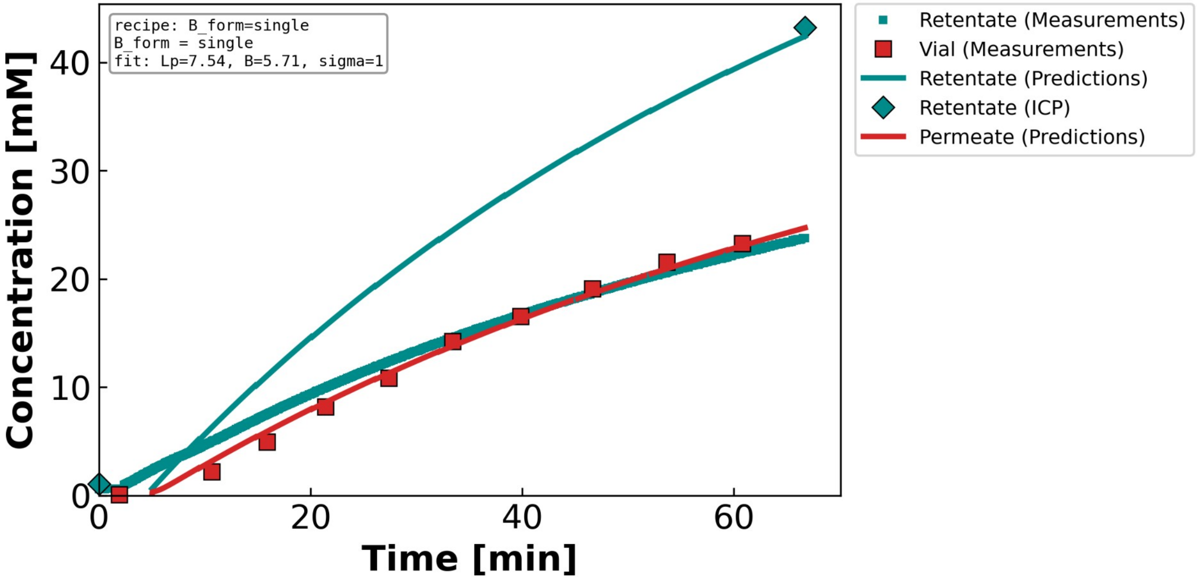
Backup — Figure 2. Measured and model-predicted time profiles for a representative…



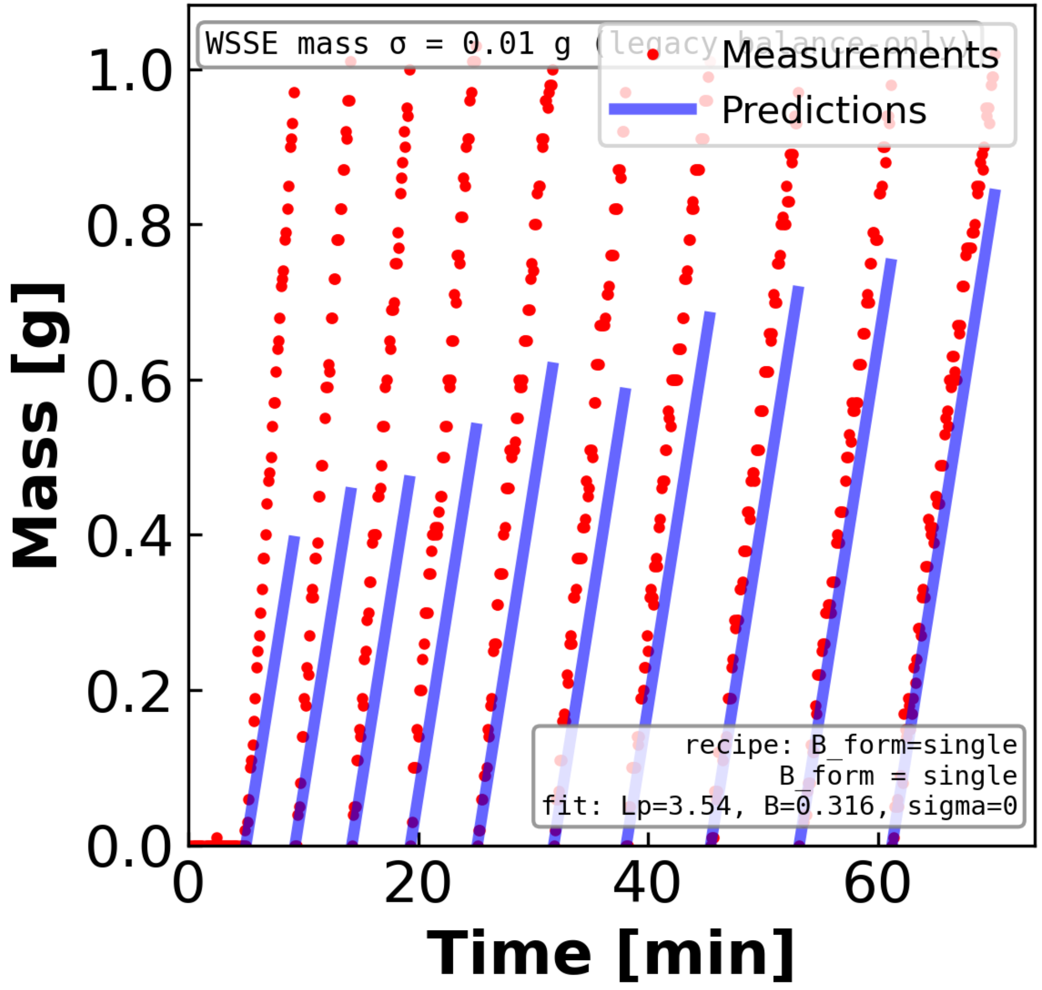


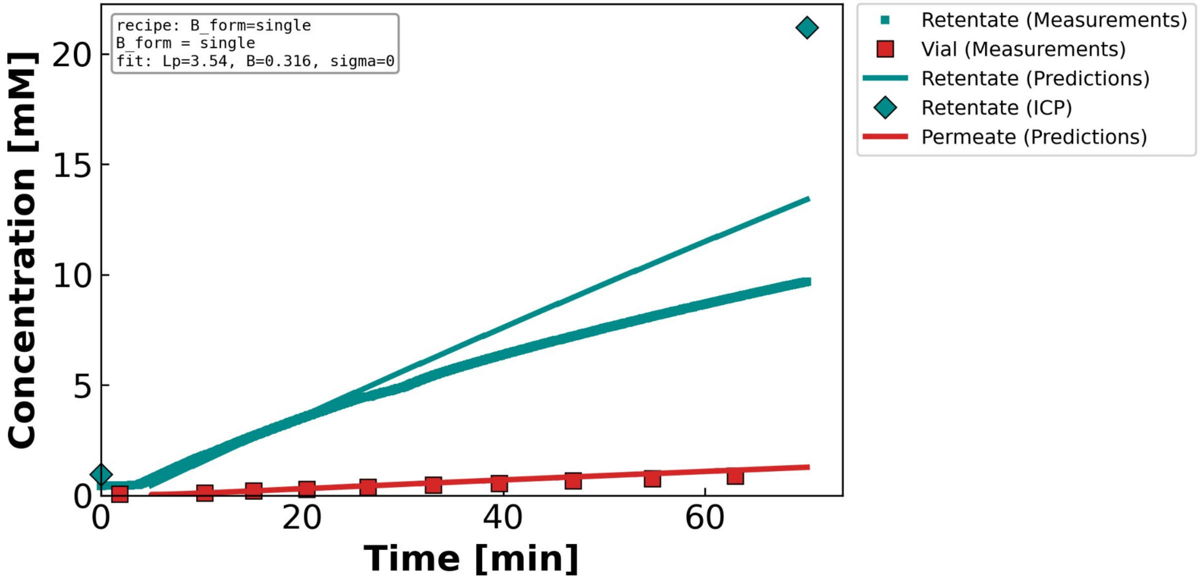
Backup — Figure 3. Measured and model-predicted time profiles for a representative…



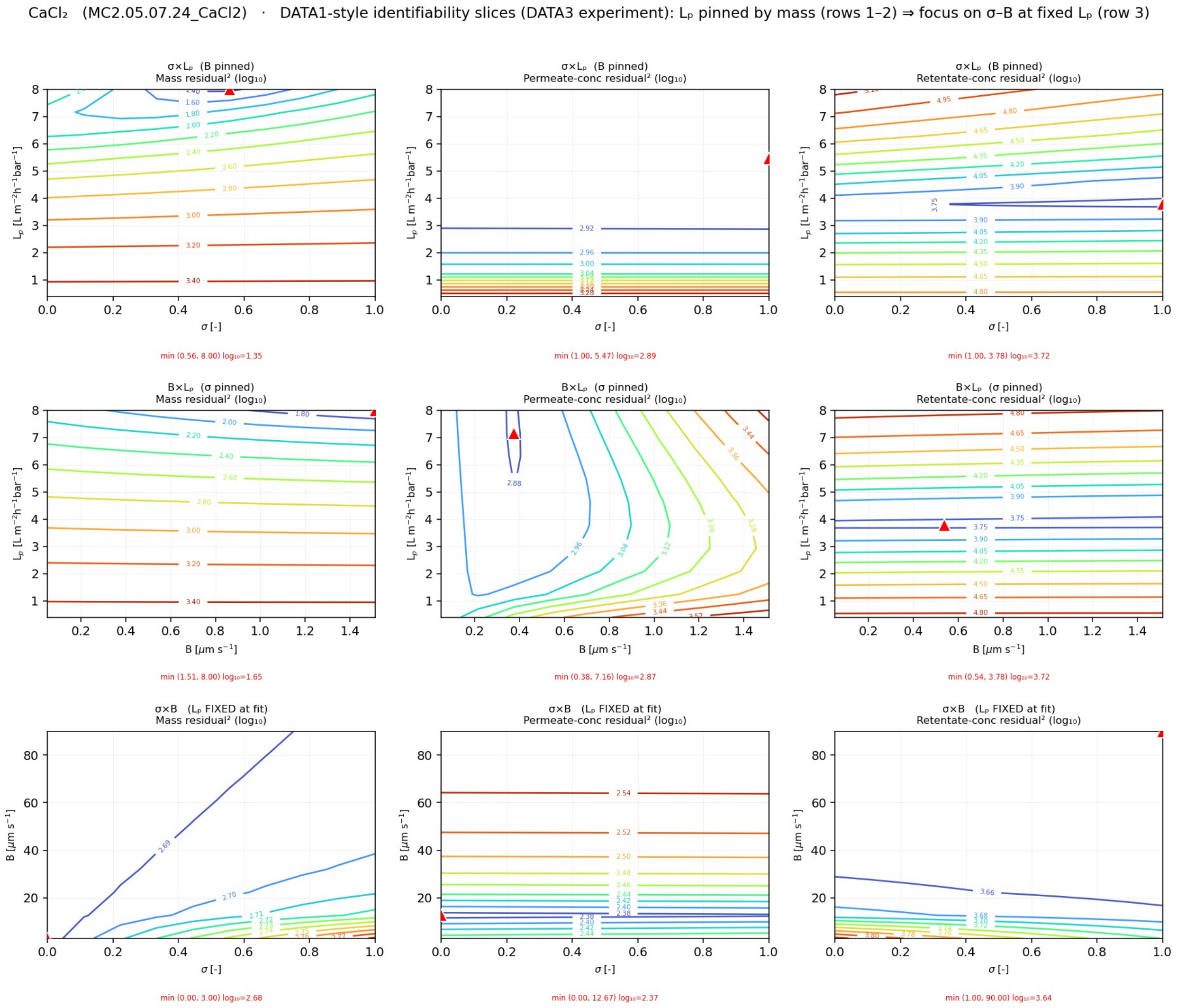


Backup — Figure 4. Measured and model-predicted time profiles for a representative…

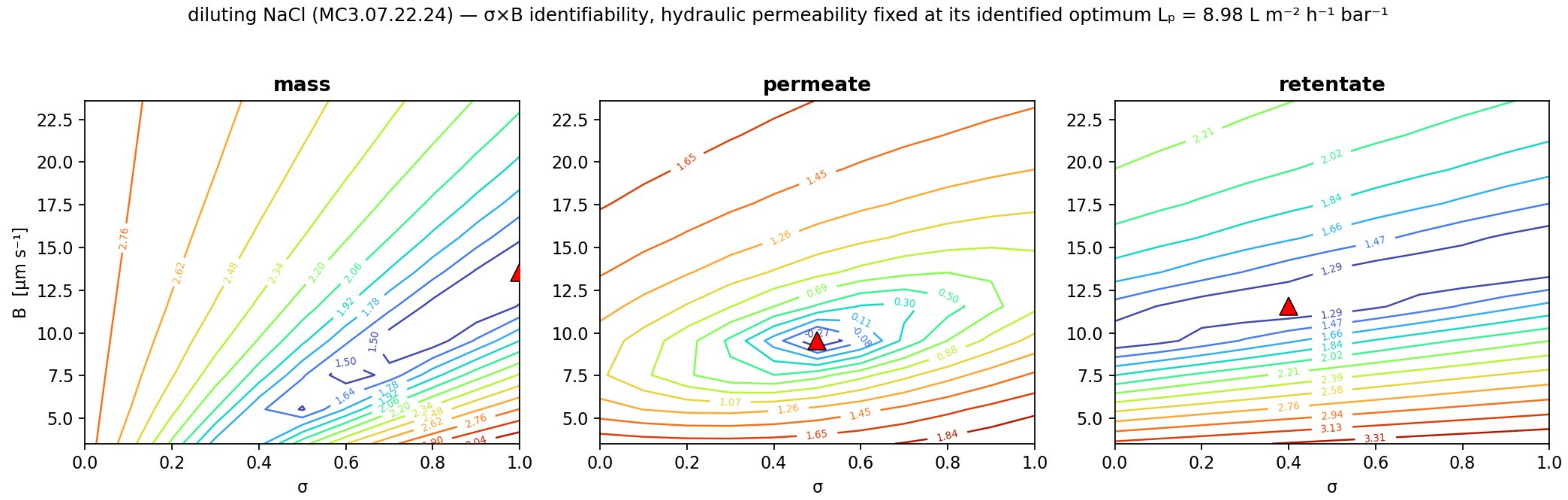




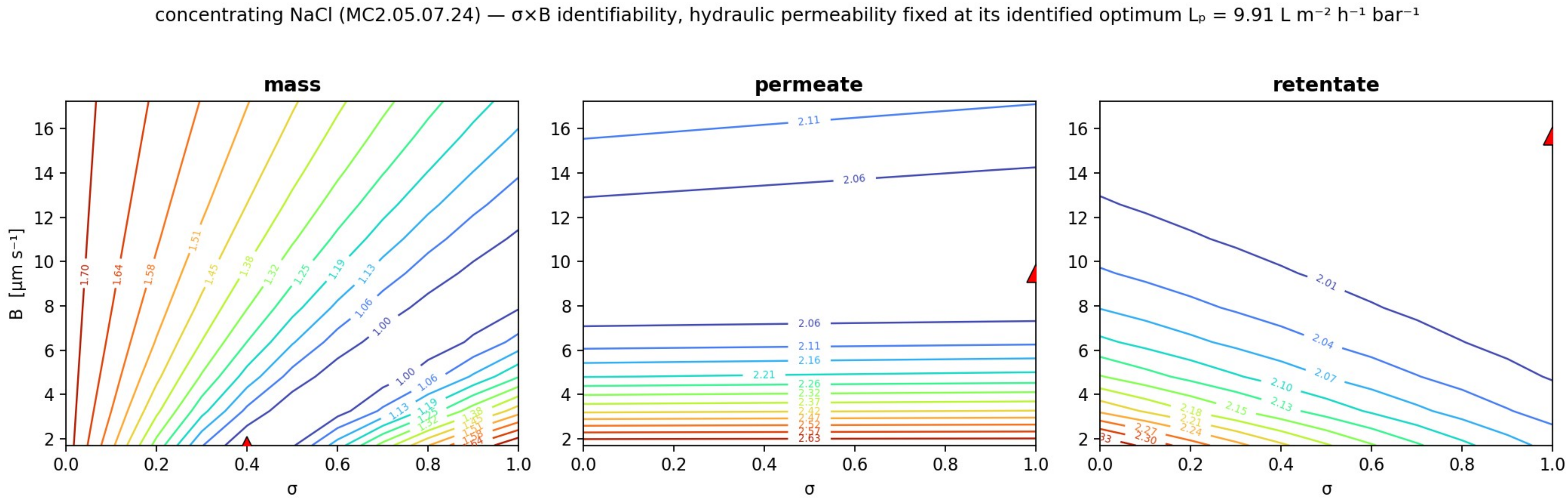
Backup — Figure 5. Log-transformed residual-squared contours for a representative…



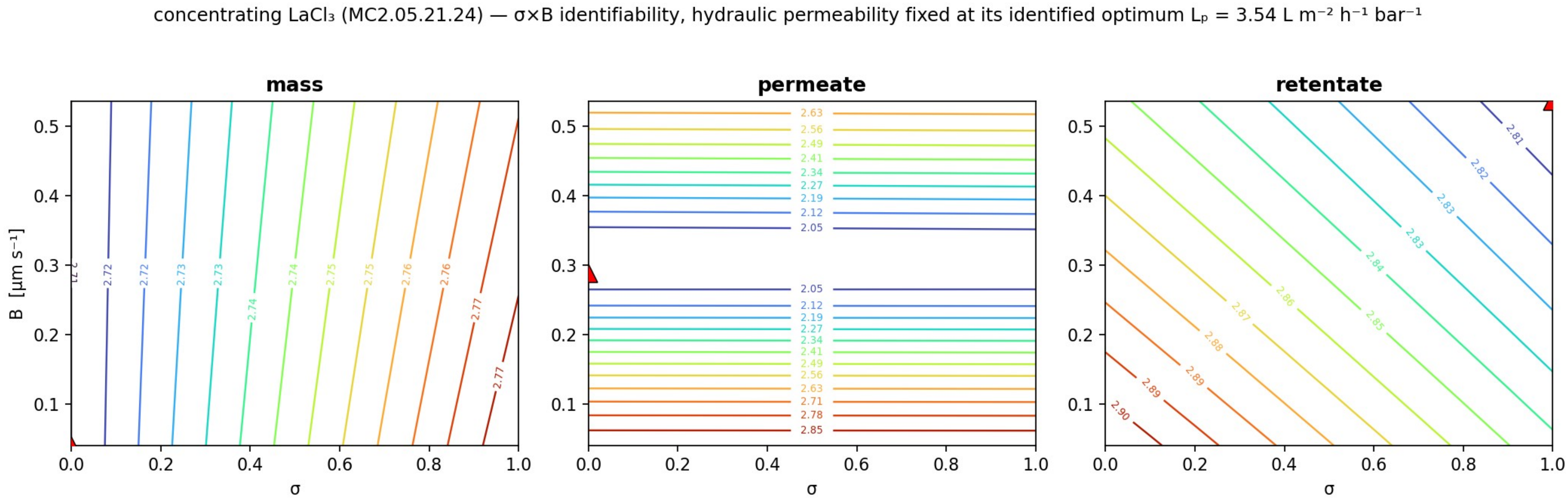
Backup — Figure 6. Identifiability of the reflection coefficient σ and the solute…



Backup — Figure 7. Identifiability of the reflection coefficient σ and the solute…

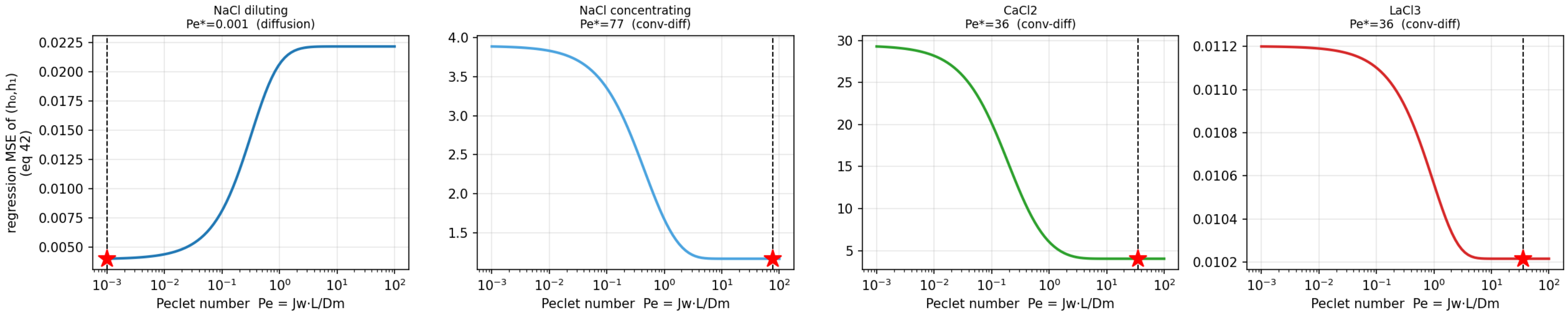


Backup — Figure 9. Identifiability of the reflection coefficient σ and the solute…

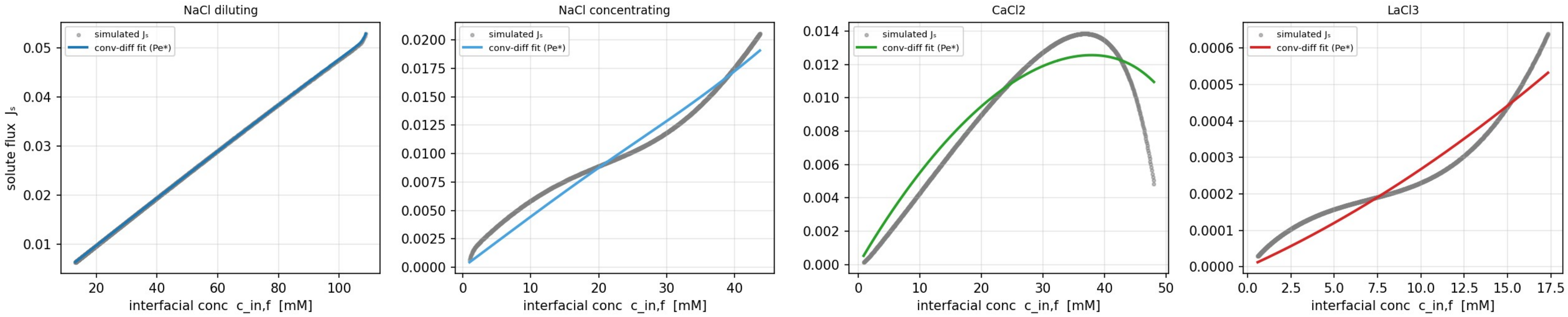


Backup — Figure 10. (Left) Mean squared error of the convection–diffusion…

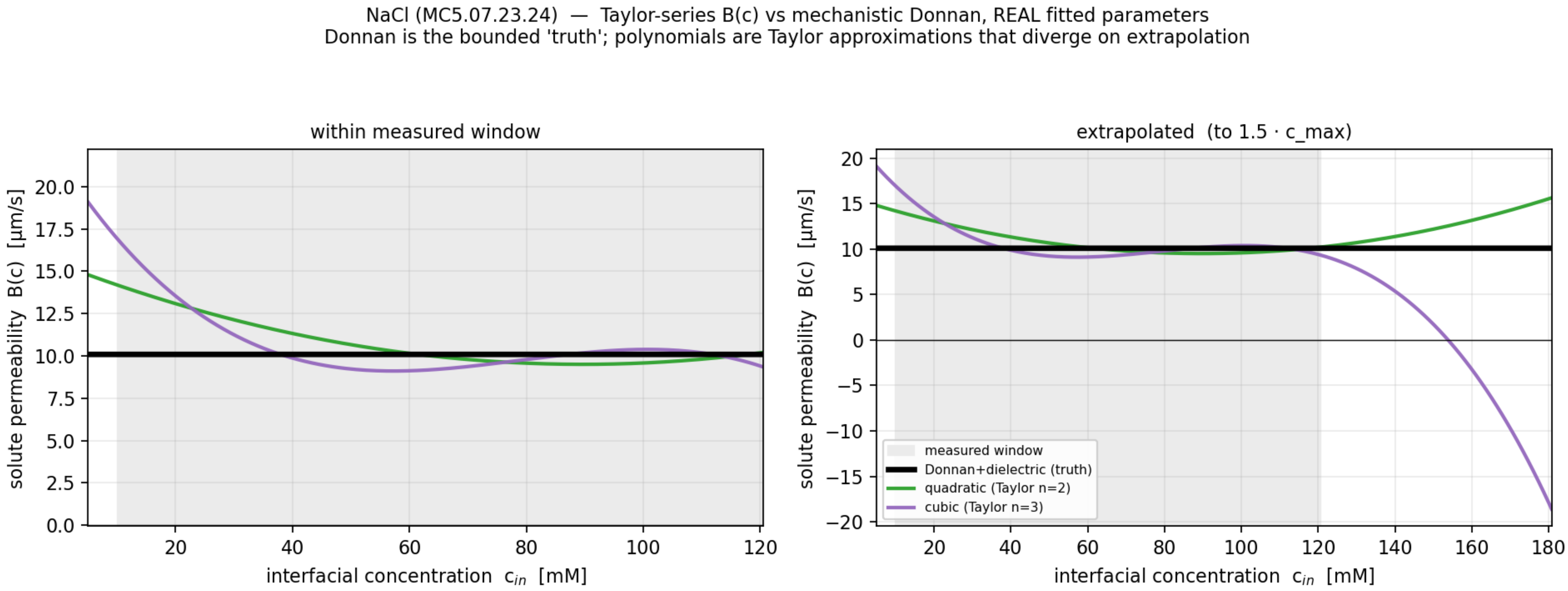
DATA3 Peclet analysis (≙ DATA2 Fig 8A): regression MSE vs Pe — a minimum at Pe≳1 means convection+diffusion, i.e. B is concentration-dependent (not the Pe→0 constant-B limit)



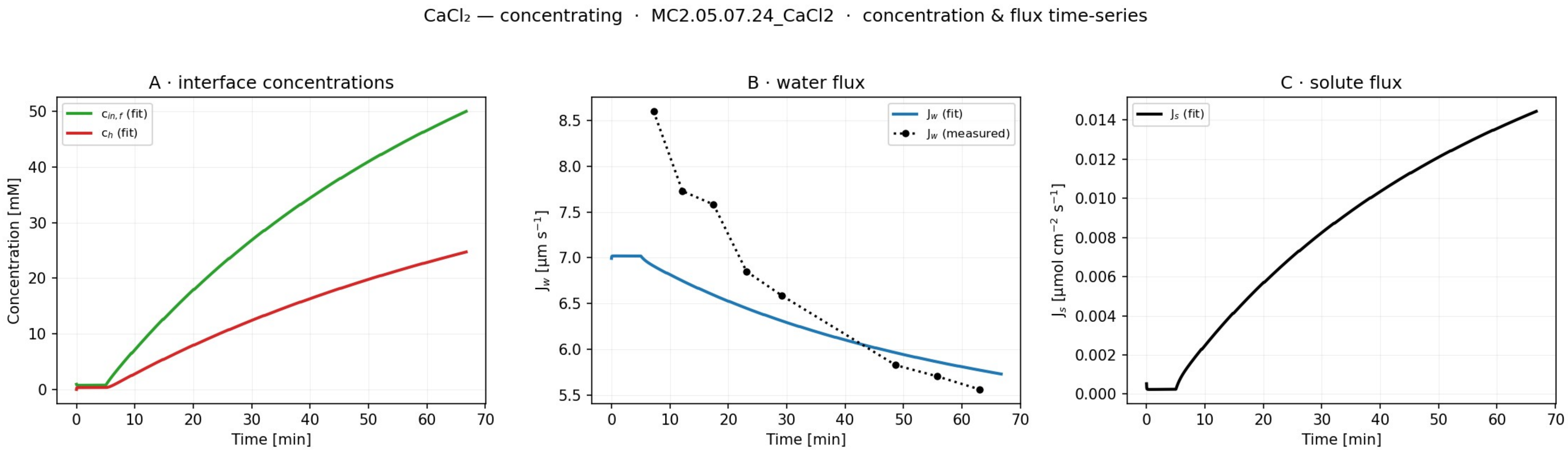
DATA3 (≙ DATA2 Fig 7C): simulated J_s vs the exact convection-diffusion solution at Pe*



Backup — Figure 15. Solute permeability B as a function of interfacial concentration for…



Backup — Figure 16. Forward-simulated membrane-interface concentration c_{in},f and bulk…



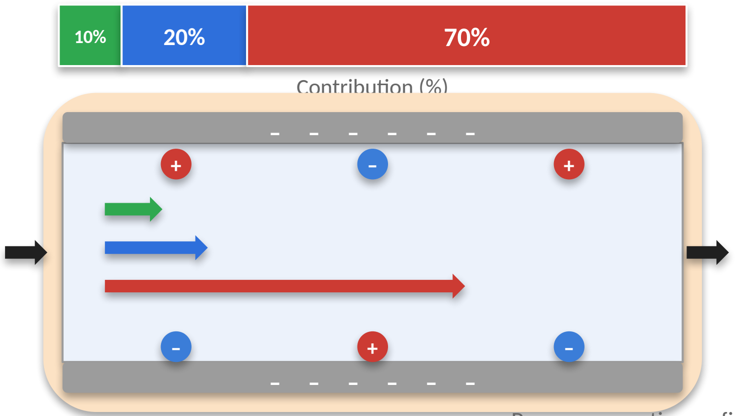
Backup — Shifting Transport Dominance (Extended Nernst–Planck)

$$J_i = v_s \cdot C_i \cdot K_{i,v} - D_i \cdot \nabla C_i - z_i \cdot C_i \cdot D_i \cdot (F/RT) \cdot \nabla \psi$$

Convection Diffusion Electromigration

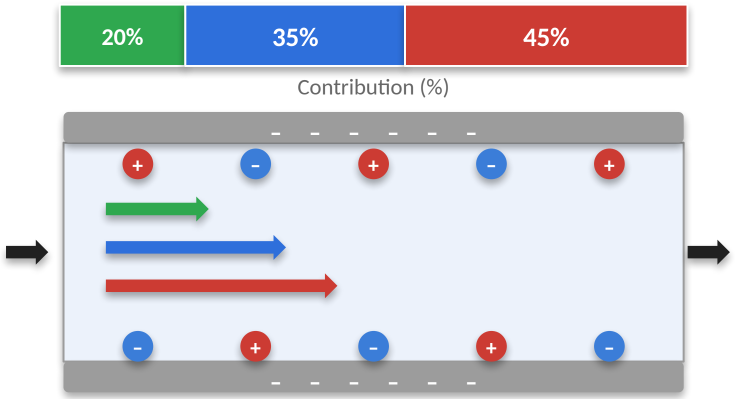
Low Molarity (~0.01–0.1 M)

Contributions to flux (J_i)



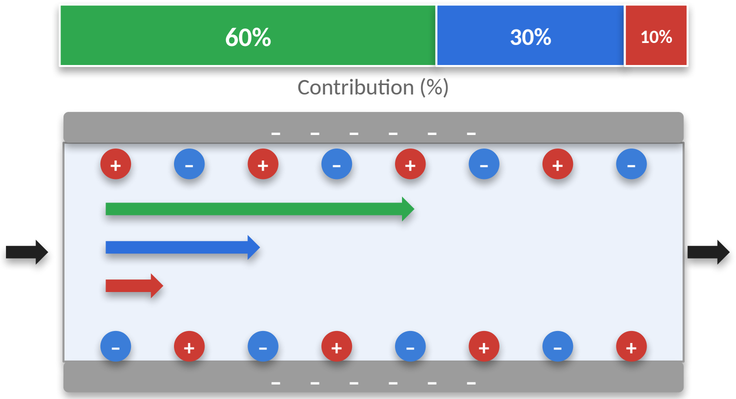
Moderate Molarity (~0.1–0.5 M)

Contributions to flux (J_i)



High Molarity (~0.5–2+ M)

Contributions to flux (J_i)



Pore cross-section — fixed (−) wall charges, mobile ions (+/−), flux arrows sized by contribution; flow is Feed → Permeate (left → right)

Low molarity:
- **Electromigration** (dominant)
- **Diffusion** (secondary)
- **Convection** (minor)

Moderate molarity:
- **Convection** (increasing)
- **Diffusion** (comparable)
- **Electromigration** (significant)

High molarity:
- **Convection** (dominant)
- **Diffusion** (secondary)
- **Electromigration** (minor)

ψ

Role of fixed negative charges & the Donnan potential. Negatively charged pore walls repel co-ions and attract counter-ions, creating a Donnan potential ψD that drives electromigration. At low molarity the low ionic strength magnifies ψD, making electromigration the primary driver; as molarity rises, charge screening by abundant ions shrinks ψD, so convection increasingly governs transport.